

Roll Number _____

Group: _____



Thapar Institute of Engineering & Technology, Patiala
(Deemed to be University)

Department of Electronics & Communication Engineering
(MST Written Test)

M.Tech(VLSI)

Date- 7th March, 2025

Time: 02 Hours, MM: 30

PVL333: System on Chip

Name of Faculty: Dr. Shalabh Kumar Mishra

NOTE: All questions carry equal weightage.
Assume any missing information.

Q. No.	QUESTIONS	CO	BL
1.	What is the Top-down System-on-Chip (SoC) design flow? Discuss its significance. Explain the various stages involved in the Top-down SoC design flow [6 Marks]	CO1	L2, L3
2.	Discuss the different metrics used in SoC design. What is the significance of the Power-Delay Product (PDP) in SoC design? [6 Marks]	CO1	L2, L3,
3.	What is the primary distinction between ASIC (Application-Specific Integrated Circuit) and SoC (System-on-Chip)? Additionally, explain the role of EDA tools in the ASIC design process. [2+4Marks]	CO2	L3
4.	Describe the typical flow of ASIC design, outlining each stage of the process, from initial specification to final manufacturing. [6 Marks]	CO2	L2, L3,
5.	(a) Discuss the key components involved in Front-end Chip design. (b) Write short note on following: (i) Intellectual Property (IP) cores (ii) Network on Chip (NoC) [3+3 Marks]	CO1 CO2 CO3	L2, L3,

***** All the Best*****